Independent projectile directions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why horizontal and vertical projectile motion share time but evolve differently.

SUCCESS CRITERIA - I can...

1. Identify horizontal and vertical velocity components.
2. Use the same elapsed time for both axes.
3. State the negligible-air-resistance assumption.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Outside a spherical source, field at r is 75 N/kg. Find field at 5 r and the force on 0.7 kg there.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $a_x = 0$; $a_y = -g$; v_x = constant.
Upward positive; g = 10 m/s². Ideal near-Earth motion with negligible air resistance.

Worked reasoning: A ball launches horizontally at 3 m/s and falls for 2 s. Horizontal travel is 3×2=6 m. With g=10 m/s² and no initial vertical velocity, its final vertical velocity is 20 m/s down.

Your turn: Draw separate horizontal and vertical velocity arrows at launch and after 2 s.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

03 Read and interpret the model

One ball is dropped and another launched horizontally from the same height at the same instant. Compare fall times and explain.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

04 Apply the physics

A ball rolls horizontally off a platform at 8 m/s. It lands 1.5 s later. Label the model and find horizontal displacement, vertical displacement and both final velocity components. Take upward as positive, use g = 10 m/s² and ignore air resistance.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

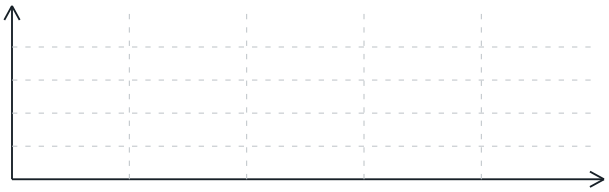
A student says a larger horizontal speed holds the ball up longer. Explain which force or acceleration would be needed for that claim.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

For $x_i=4\pm1$ m, $v=6\pm0.5$ m/s and exact $t=8$ s, find the central final position and conservative interval. Name a specific check of your method.
Label the measured positions and the range each allows.

Position and uncertainty (m)



Time (s)

Marzano check - circle: 1 2 3

07 Exit check

A projectile's velocity is (6,-8) m/s. Find speed. What information is still needed to determine its position? Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.

Separate horizontal and vertical motion

Label x and y; state positive directions.

Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A projectile has initial velocity (6, 8) m/s; use upward positive and g=10 m/s². Find velocity after 1 s and explain whether its horizontal motion depends on its vertical velocity. Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3